MZ2POL-03: Assessment and Migration Planning

> **Series:** MZ2POL — Management Zone Migration: Segments, Policies, Workflows |
> **Notebook:** 4 of 17 | **Created:** December 2025 | **Last Updated:** 10/14/2026

Overview

This notebook guides you through a comprehensive assessment of your current Management Zone implementation and helps you create a detailed migration plan to the new Policies, Boundaries, and Segments model.

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Prerequisites

- Completed MZ2POL-01 and MZ2POL-02
- Admin access to Dynatrace environment

- Access to Account Management
- List of current user groups and their MZ assignments

Learning Objectives

By the end of this notebook, you will:
1. Have a complete inventory of your Management Zones
2. Understand the mapping from MZs to new constructs
3. Have a prioritized migration plan
4. Know the risks and mitigation strategies

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1. Management Zone Inventory

> ⚠️ **Important:** Management Zone configurations **cannot be queried via DQL**.
>
> **Options for MZ inventory:**
> 1. **Dynatrace UI (Recommended):** Settings → Management Zones
> 2. **SDK Analysis:** See **MZ2POL-00: SDK Management Zone Analysis Tool** for comprehensive analysis including rule patterns, coverage metrics, and migration readiness.

Entity Distribution Across MZs

> ⚠️ **Limitation:** DQL queries against `managementZones` may not accurately reflect the full MZ entity distribution due to how MZ rules propagate. For accurate counts, use the **classic Dynatrace UI** which shows entity counts directly on each Management Zone configuration.

The query below provides an approximation of entity distribution:

```
// Service distribution by Management Zone
fetch dt.entity.service
| expand mz = managementZones
| summarize serviceCount = count(), by:{managementZone = mz}
| sort serviceCount desc
| limit 25

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - management zones have NO Smartscape node equivalent; they are what this migration
//     replaces with segments and policies.
```

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2. MZ Pattern Classification

Categorize Your Management Zones

Group your MZs by their primary purpose:

Category 1: Team/Ownership Based

- **Pattern**: MZs named after teams (e.g., "Team-Frontend", "Platform-Team")
- **Migration**: → Security Context + Policies + Boundaries
- **Priority**: High (affects user access)

Category 2: Environment Based

- **Pattern**: MZs for environments (e.g., "Production", "Staging", "Development")
- **Migration**: → A record field that carries the environment. Boundaries have no environment attribute of their own: for access, put the stage into the `dt.security_context` value (`prod/payments`) or use a built-in field the stage already maps to (`dt.host_group.id` , `k8s.namespace.name`); for filtering only, a primary tag such as `primary_tags.stage` in a segment
- **Priority**: High (critical for access separation)

Category 3: Geographic/Region Based

- **Pattern**: MZs for regions (e.g., "NA-East", "EU-West", "APAC")
- **Migration**: → Segments with cloud region/location filters
- **Priority**: Medium (primarily filtering)

Category 4: Application/Service Based

- **Pattern**: MZs for applications (e.g., "Payment-System", "User-Portal")
- **Migration**: → Segments with service/application filters
- **Priority**: Medium (primarily filtering)

Category 5: Infrastructure Based

- **Pattern**: MZs for infrastructure (e.g., "Kubernetes", "Legacy-VMs")
- **Migration**: → Segments with infrastructure type filters
- **Priority**: Low (usually informational)

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3. Security Context and Bucket Planning

Design Your Security Context Strategy

Security context drives record-level access control. It is **one field**, read by every IAM record condition, so design **one value per owner** and write it onto every record that owner produces — not a different dimension per entity type. If services carried `team-*` and hosts `env-*`, a team boundary would miss the host's logs and an environment boundary would miss the service's spans.

Dimension	Where it is written	Example value
Access (team, plus stage where access depends on it)	`dt.security_context` on every record the team produces	`payments`, or `prod/payments`
Filtering and routing	`primary_tags.*`, in the same enrichment step from the same source	`primary_tags.team = payments`, `primary_tags.stage = prod`

Write both together so they cannot drift apart — **MZ2POL-10 §4** (*One Value, Every Job*). Structured values such as `prod/payments` are covered in **MZ2POL-04 §4**.

Design Your Bucket Strategy

> ⚠️ **Critical:** Plan buckets carefully - names are immutable and data cannot be moved between buckets.

A zone that restricted access does **not** become a bucket by default. Buckets are *"not intended as the primary mechanism for securing or segmenting data access"*; team access is a boundary on `dt.security_context` (**MZ2POL-12: IAM Reference Architecture**, **ORGNZ-06: Security Context**). Create buckets for what they are for — retention, cost allocation and query optimization — and for a specific compliance case that needs a hard exclusion (**ORGNZ-05: Bucket-Level Access Control**).

Bucket Decisions:

Decision	Options	Considerations
Partitioning dimension	Retention period, cost owner, query scope, compliance case	What differs in how the data must be stored or billed?
Naming convention	`{purpose}_{datatype}_{retention}`	Name the reason the bucket exists, plus type and retention
Number of buckets	One per distinct retention / cost / compliance need	Fewer is simpler — access does not need one per team

> **Sources:** Partition data (DT docs) — buckets are *"best used for organizing data by retention time, cost allocation, or query optimization."*

Bucket Limitations to Consider:

Limitation	Impact on Planning
One data type per bucket	Need separate buckets for logs, spans, metrics

Limitation	Impact on Planning
Maximum custom buckets per environment: 250 by default from SaaS 1.346 (staged tenant rollout from 08/25/2026); 80 before it	Plan for scale within your tenant's limit. *Organize data* (updated 09/22/2026) and the 1.346 release note state 250; *Partition data* (updated 06/17/2026) and the upgrade guide's *Data partitioning* page (published 08/24/2026) still state 80 — check which default your tenant has before planning to either number
~1 TB/day optimal per bucket	Best query performance
1-3 TB/day acceptable	Shorter query window per default 500 GB scan
No documented hard cap per bucket	Dynatrace sizing guidance on where to split differs between pages (Partition data (DT docs), Logs bucket strategy (DT docs))
Names immutable	Choose naming convention carefully upfront
No data migration	Cannot consolidate or split buckets later

> **Sources:** Organize data (DT docs) — *"The default limit per environment is 250 custom buckets."*; SaaS 1.346 (DT docs) — *"Dynatrace now supports up to 250 custom Grail buckets by default"*; Partition data (DT docs) — *"For each Dynatrace environment, you can configure up to 80 custom buckets by default."*; Data partitioning — upgrade best practices (DT docs) — *"The default bucket limit per environment is 80 buckets"*.

Default query setting (affects bucket sizing): `fetch` reads at most 500 GB of uncompressed data unless the query sets `scanLimitGBytes` — *"The default value is 500GB unless specified otherwise."* (Data source commands (DT docs))

Mapping MZs to Buckets

For each MZ, determine if bucket partitioning is needed — usually it is not:

MZ Pattern	Bucket Recommendation
Team-based MZ, including strict team isolation	**Not needed** — a security-context value per team and one boundary per team (MZ2POL-12)
Environment MZ (Prod/Dev separation)	Only if the environments need different retention; access uses the stage in the security-context value

MZ Pattern	Bucket Recommendation
Compliance MZ (PCI, HIPAA)	Consider a compliance bucket where retention or a hard exclusion (`DENY` on the bucket) is required (ORGNZ-05)
Regional MZ (filtering only)	Usually **not needed** - use Segments instead
Application MZ (informational)	Usually **not needed** - use Segments instead

> **For comprehensive bucket guidance**, see **ORGNZ-03: Bucket Strategy and Design**.

Measure Security Context on the Data

IAM record conditions test the **record**, so measure coverage on logs, spans and events — not on entities. Entity coverage misleads: the classic *Grail Security Context* mapping can copy zone names onto entities, while records *"do not inherit the management zones of that entity."* On the validation tenant (10/13/2026) every service and host carried a zone-derived security context, while 41% of logs and 75% of events in the last hour carried none.

> **Sources:** Grant access to entities with security context (DT docs) — *"Logs, spans, metrics, and events powered by Grail that are sent from an entity do not inherit the management zones of that entity."*

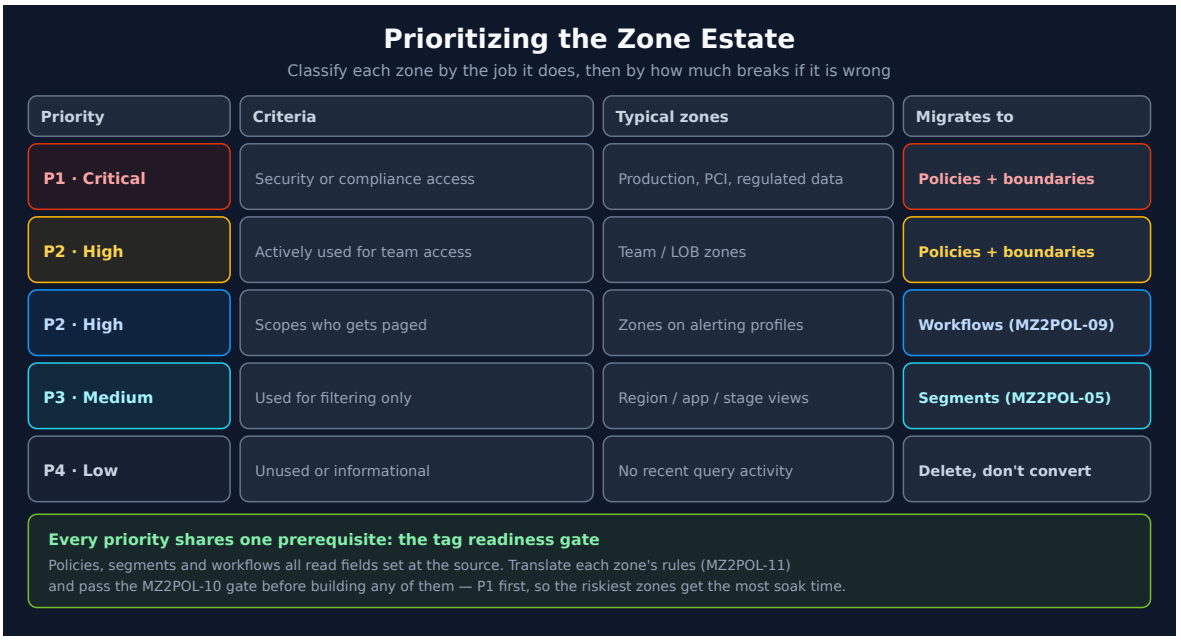
First, where the uncovered records come from — grouped by the producer fields an enrichment rule can key on (MZ2POL-10). Then the per-table share.

```
// Logs without a security context, by producer — the tagging targets.
fetch logs, from:-1h
| filter isNull(dt.security_context)
| summarize {records = count()}, by:{dt.host_group.id, k8s.namespace.name,
host.name}
| sort records desc
| limit 50
```

```
// Security-context coverage on the DATA, per table (same check as MZ2POL-00
§5 and MZ2POL-06 §3).
// Events carry dt.security_context as an array, so an empty array also counts
as missing.
fetch logs, from:-1h
| summarize {records = count(), without_sc =
countIf(isNull(dt.security_context))}
| fieldsAdd table = "logs"
| append [fetch spans, from:-1h | summarize {records = count(), without_sc =
countIf(isNull(dt.security_context))} | fieldsAdd table = "spans"]
| append [fetch events, from:-1h | summarize {records = count(), without_sc =
countIf(isNull(dt.security_context) or arraySize(dt.security_context) == 0)} |
fieldsAdd table = "events"]
| fieldsAdd without_pct = round(100.0 * without_sc / records, decimals: 1)
| fields table, records, without_sc, without_pct
```

4. Migration Priority Matrix

Prioritize Based on Impact and Complexity



Priority	Criteria	Examples
P1 - Critical	Security/compliance requirements	Production access control
P2 - High	Actively used for access control	Team-based MZs

Priority	Criteria	Examples
P3 - Medium	Used for filtering only	Region/app MZs
P4 - Low	Informational/rarely used	Legacy MZs

Risk Assessment

Risk	Likelihood	Impact	Mitigation
Users lose access	Medium	High	Parallel running period
Wrong data visible	Medium	High	Thorough testing
Alert routing breaks	High	Medium	Update alerts before cutover
Dashboard filters fail	Medium	Low	Pre-update dashboards

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5. Migration Timeline Template

Recommended Phases

Phase 1: Preparation

- [ ] Complete MZ inventory
- [ ] Document current user access patterns
- [ ] Create migration mapping for all MZs
- [ ] Design security context strategy
- [ ] **Design bucket strategy** (only for retention, cost or a compliance case)
- [ ] Identify dependencies (alerts, dashboards, APIs)

Phase 2: Foundation - Buckets and Security Context

- [ ] **Create Grail buckets** (if applicable)
- [ ] **Configure OpenPipeline routing** to buckets
- [ ] Write security context onto the data at the source (MZ2POL-10) and re-run the coverage check in §3

- [ ] Create user groups in Account Management
- [ ] Set up IAM policy structure
- [ ] Create boundary definitions

> ⚠️ **Bucket Order:** Create buckets and configure pipeline routing BEFORE migrating policies. Data must flow to correct buckets first.

Phase 3: Segments

- [ ] Create segments for data filtering
- [ ] Test segment functionality
- [ ] Update dashboards to use segments
- [ ] Validate filtering behavior

Phase 4: Policy Assignment

- [ ] Create bucket-based policies (compliance cases only)
- [ ] Bind policies to groups
- [ ] Apply boundaries to policy bindings
- [ ] Test user access (parallel with MZs)
- [ ] Document access verification results

Phase 5: Cutover

- [ ] Rebuild MZ-scoped alerting profiles as problem-triggered workflows (affected-entity tags + DQL matcher) — **not** as segments
- [ ] Migrate API integrations
- [ ] Remove MZ-based access (RBAC)
- [ ] Monitor for access issues

Phase 6: Cleanup

- [ ] Verify no MZ dependencies remain
- [ ] Archive MZ configurations
- [ ] Remove unused MZs
- [ ] Update documentation

Bucket Migration Checklist

If a retention, cost or compliance case needs dedicated buckets:

- [ ] Bucket naming convention finalized
- [ ] Buckets created for each data type (logs, spans, metrics)
- [ ] OpenPipeline routing rules configured
- [ ] Data flowing to correct buckets verified
- [ ] Bucket-specific policies created
- [ ] Policies include `WHERE storage:bucket-name = "..."` conditions
- [ ] Boundaries include `storage:bucket-name IN (...)` restrictions

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6. Dependencies Checklist

What References Management Zones?

Before migrating, identify all MZ dependencies. This checklist names the categories; MZ2POL-15 finds the actual objects in settings, classic configuration, the account and code, and builds the dependency matrix:

Alerting

- [ ] Alerting profiles using MZ filters
- [ ] Problem notifications scoped to MZs
- [ ] Metric events with MZ dimensions

Dashboards & Reports

- [ ] Dashboards with MZ filters
- [ ] Reports scoped to MZs
- [ ] SLOs using MZ-filtered data

Integrations

- [ ] API calls with MZ parameters
- [ ] Automation workflows using MZ scopes

- [ ] Third-party integrations filtering by MZ

Access Control

- [ ] User groups with MZ assignments
- [ ] RBAC roles using MZ restrictions

Summary

In this notebook, you:

1. **Inventoried** all Management Zones with DQL queries
2. **Classified** MZs by pattern (team, environment, region, etc.)
3. **Created** migration mapping templates
4. **Planned** security context strategy
5. **Prioritized** migration based on risk and impact
6. **Identified** dependencies to address

Deliverables Checklist

- [ ] Complete MZ inventory spreadsheet
- [ ] Migration mapping document for each MZ
- [ ] Security context assignment plan
- [ ] Prioritized migration schedule
- [ ] Risk assessment and mitigation plan
- [ ] Dependencies checklist

Next Steps

First, the tagging foundation. Every replacement for a management zone reads a field on the data, so the tags have to be in place before you build on them:
- **MZ2POL-10: Tagging at Source** — the model change, the enrichment ladder, and the readiness gate

- **MZ2POL-11: Translating MZ Rules into Source Attributes** — the rule-by-rule catalog that turns this inventory into target fields
- **MZ2POL-12: IAM Reference Architecture** — how the access-control zones become groups, default policies and one boundary per team, and the boundary pitfalls that widen access
- **MZ2POL-14 [WORKSHOP]: Zone Disposition** — run the decisions in this notebook with the zone owners; **MZ2POL-95 [LAB]** is the worked example on a sample estate

Then continue to **MZ2POL-04: Policies and Boundaries** to:
- Learn policy syntax and structure
- Create custom policies
- Understand default policy capabilities

Additional Resources

- Management Zones Documentation
- Upgrade from RBAC to IAM Policies
- Grant Access to Entities with Security Context

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